

QUESTION:

1. Sample Java program to check whether the given number is prime palindrome or not.
CO1.

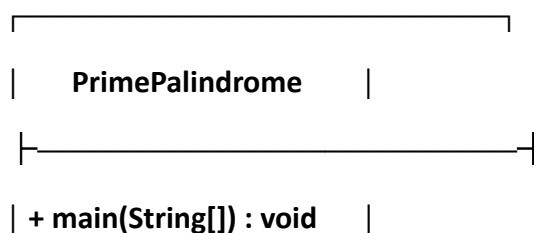
Steps for Execution

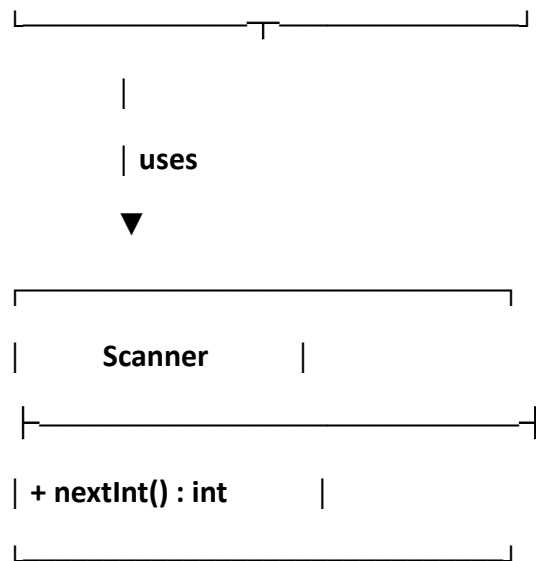
- 1.Open IntelliJ IDEA / Eclipse / VS Code.
- 2.Create a Java project.
- 3.Create a package named javaprograms.
- 4.Create a class named PrimePalindrome.
- 5.Enter the given code and save it.
- 6.Run the PrimePalindrome class.
- 7.Enter a number when prompted.
- 8.The program checks whether the number is both prime and palindrome.
- 9.Display the result.

Algorithm

- 1.Start.
- 2.Create a Scanner object.
- 3.Read the number n.
- 4.Reverse the number using a while loop.
- 5.Check whether the reversed number is equal to the original number.
- 6.Count the number of factors of n using a for loop.
- 7.If the number has exactly 2 factors and is a palindrome, it is a Prime Palindrome.
- 8.Otherwise, display not a Prime Palindrome.
- 9.Stop.

UML DIAGRAM:





:

CODE:

```

package javaprograms;
import java.util.Scanner;

public class PrimePalindrome {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        int rev = 0, temp = n, count = 0;

        // Check palindrome
        while (temp > 0) {
            rev = rev * 10 + temp % 10;
            temp = temp / 10;
        }

        // Check prime
        for (int i = 1; i <= n; i++) {
            if (n % i == 0)
                count++;
        }

        if (rev == n && count == 2)
  
```

```
        System.out.println(n + " is a Prime Palindrome");
    else
        System.out.println(n + " is not a Prime Palindrome");
    }
}
```

OUTPUT:

Enter a number: 131

131 is a Prime Palindrome

QUESTION:

2. Write a java program which reads marks of five subjects through command line and print the total and average.

Steps for Execution

- 1.Open IntelliJ IDEA / Eclipse / VS Code.
- 2.Create a Java project.
- 3.Create a package named javaprograms.
- 4.Create a class named MarksCalculation.
- 5.Enter the given code and save it.
- 6.Run the MarksCalculation class.
- 7.Enter the five marks when prompted.
- 8.The total and average will be displayed.

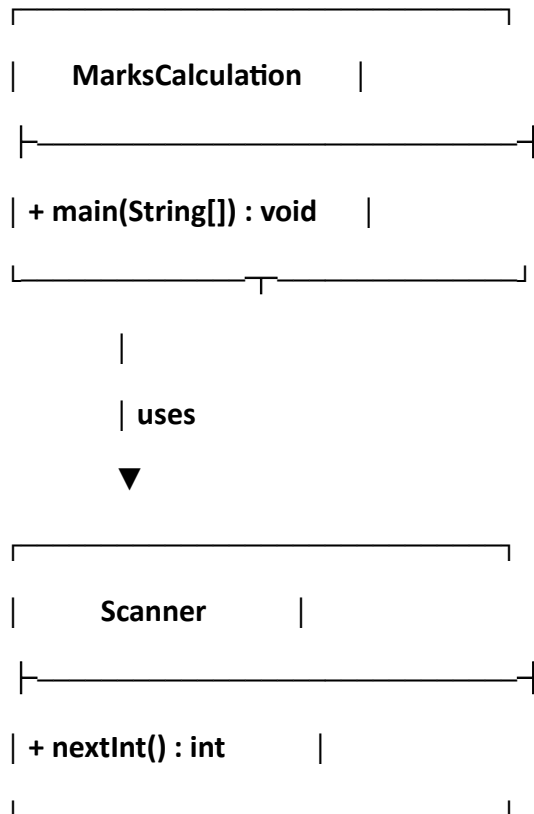
Algorithm

- 1.Start.
- 2.Create a Scanner object to read input.
- 3.Read five marks: m1, m2, m3, m4, m5.
- 4.Calculate total:
5. $\text{total} = m1 + m2 + m3 + m4 + m5$
- 6.Calculate average:
7. $\text{average} = \text{total} / 5.0$
- 8.Display the total.

9.Display the average.

10.Stop.

UML DIAGRAM:



CODE:

```
package javaprograms;

import java.util.Scanner;

public class MarksCalculation {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Mark 1: ");

        int m1 = sc.nextInt();

        System.out.print("Enter Mark 2: ");
```

```
int m2 = sc.nextInt();

System.out.print("Enter Mark 3: ");
int m3 = sc.nextInt();

System.out.print("Enter Mark 4: ");
int m4 = sc.nextInt();

System.out.print("Enter Mark 5: ");
int m5 = sc.nextInt();

int total = m1 + m2 + m3 + m4 + m5;
double average = total / 5.0;

System.out.println("Total = " + total);
System.out.println("Average = " + average);
}
}
```

OUTPUT:

Enter Mark 1: 80

Enter Mark 2: 75

Enter Mark 3: 90

Enter Mark 4: 85

Enter Mark 5: 70

Total = 400

Average = 80.0